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OPTICAL BRANCHING CABLE SET

Abstract

An optical branching cable set includes a plurality of optical branching cables, and a branching cord bag. At least two of the optical branching cables each include a main cable including a plurality of optical cords each including at least one optical fiber, a tensile strength member, and a tape that forms an outermost layer. A branching cord portion including at least one optical fiber that is led out from the optical cords of the main cable to an outside of the outermost layer. The branching cord bag is configured to collectively accommodate a plurality of branching cord portions provided in the plurality of optical branching cables.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2024-081993 filed on May 20, 2024, Japanese Patent Application No. 2024-032396 filed on Mar. 4, 2024, and Japanese Patent Application No. 2024-032397 filed on Mar. 4, 2024, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an optical branching cable set.

BACKGROUND ART

[0003] JP2023-083096A discloses an optical branching cable that connects a server rack group including a plurality of server racks and a distribution board. The optical branching cable includes a main cable and branching cords that branch from the main cable and that are connected to the server racks.

[0004] The arrangement of the server racks in the server rack group varies depending on the intended use. Therefore, the optical branching cable is manufactured such that the positions of the branching portions, where the main cable and the branching cord portion are branched, correspond to the arrangement of the server racks. The branching cord portion needs to be sufficiently protected at the time of laying, and it is also important that the optical branching cable be easy to manufacture.

SUMMARY OF INVENTION

[0005] Aspect of non-limiting embodiments of the present disclosure relates to provide an optical branching cable set that achieves both protection of a branching cord portion and ease of manufacture.

[0006] Aspects of certain non-limiting embodiments of the present disclosure address the features discussed above and/or other features not described above. However, aspects of the non-limiting embodiments are not required to address the above features, and aspects of the non-limiting embodiments of the present disclosure may not address features described above.

[0007] According to an aspect of the present disclosure, there is provided an optical branching cable set including: [0008] a plurality of optical branching cables; and [0009] a branching cord bag, [0010] in which at least two of the optical branching cables each include: [0011] a main cable including: [0012] a plurality of optical cords each including at least one optical fiber; [0013] a tensile strength member; and [0014] a tape that forms an outermost layer; and [0015] a branching cord portion including at least one optical fiber that is led out from the optical cords of the main cable to an outside of the outermost layer, and [0016] the branching cord bag is configured to collectively accommodate a plurality of branching cord portions provided in the plurality of optical branching cables.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0017] Exemplary embodiment(s) of the present invention will be described in detail based on the

following figures, wherein:
[0018] FIG. **1** shows an optical communication system according to the present embodiment;
[0019] FIG. **2** is a schematic view of an optical branching cable;
[0020] FIG. **3** is a cross-sectional view of a main cable;
[0021] FIG. **4** is a schematic diagram of an optical branching cable set according to a first embodiment;
[0022] FIG. **5** shows a curved optical branching cable and a branching cord bag;
[0023] FIG. **6** is a cross-sectional view taken along a plane orthogonal to the axis of the main cable of the optical branching cable set;
[0024] FIG. **7** shows the branching cord bag in a first stage;
[0025] FIG. **8** shows the branching cord bag in a second stage;
[0026] FIG. **9** shows the branching cord bag in a third stage;
[0027] FIG. **10** is a perspective view of an optical branching cable set according to a third embodiment;
[0028] FIG. **11** shows the optical branching cable set and a branching cord bag according to the third embodiment; and
[0029] FIG. **12** shows an example of arrangement of branching cord portions of optical branching cables.

DESCRIPTION OF EMBODIMENTS

Description of Embodiments of Present Disclosure

[0030] First, aspects of the present disclosure will be listed and described. (1) An optical branching cable set according to an aspect of the present disclosure including: a plurality of optical branching cables; and a branching cord bag, in which at least two of the optical branching cables each include a main cable including a plurality of optical cords each including at least one optical fiber, a tensile strength member, and a tape that forms an outermost layer, and a branching cord portion including at least one optical fiber that is led out from the optical cords of the main cable to an outside of the outermost layer, and in which the branching cord bag is configured to collectively accommodate a plurality of branching cord portions provided in the plurality of optical branching cables.
[0031] According to the optical branching cable set described above, the branching cord bag can collectively accommodate the branching cord portions of the plurality of optical branching cables. Therefore, the plurality of branching cord portions are protected by the branching cord bags, the number of which is smaller than that of the optical branching cables. Accordingly, since the number of components is reduced, it is possible to provide the optical branching cable set that achieves both protection of the branching cord portion and ease of manufacture.
[0032] (2) In the optical branching cable set according to (1) described above, the branching cord bag may be made of a material having a cushioning property.
[0033] According to the optical branching cable set described above, since the branching cord bag is made of a material having the cushioning property, the branching cord portion is more effectively protected.
[0034] (3) In the optical branching cable set according to (1) or (2) described above, the branching cord bag may be fixed to the plurality of optical branching cables at two different locations along the main cable.
[0035] According to the optical branching cable set described above, the branching cord bag is fixed to the plurality of optical branching cables at two different locations along the main cable. Therefore, the optical branching cable set can be manufactured easily, and the branching cord bag can be firmly fixed to the optical branching cable.
[0036] (4) In the optical branching cable set according to any one of (1) to (3) described above, the branching cord bag may include a base portion that is sandwiched between the main cable and the branching cord portion in a state where the branching cord portion is accommodated in the branching cord bag, and a top surface portion that is located so as to cover an upper portion of the

branching cord portion in a state where the branching cord portion is accommodated in the branching cord bag, the base portion may not cover a branching portion to which the branching cord portion is led out from the main cable, and the top surface portion may have a larger dimension along an axis of the main cable than the base portion so as to cover the branching portion.

[0037] According to the optical branching cable set described above, the base portion does not cover the branching portion. Therefore, it is easy to perform the operation of accommodating the branching cord portion led out from the branching portion. The top surface portion is formed larger than the base portion so as to cover the branching portion. Therefore, the branching cord portion including the branching portion can be protected.

[0038] (5) In the optical branching cable set according to any one of (1) to (4) described above, the branching cord bag may include an inner bag configured to be accommodated inside the branching cord bag, and the inner bag may accommodate at least a part of the branching cord portions of the optical branching cables.

[0039] According to the optical branching cable set described above, since the inner bag can be accommodated in the branching cord bag and the branching cord portion can be accommodated, the branching cord portion is more effectively protected.

[0040] (6) In the optical branching cable set according to any one of (1) to (5) described above, an outer surface of the branching cord bag may be formed with an extra length that extends according to a curvature of the optical branching cable.

[0041] The optical branching cable may be curved during manufacturing when, for example, the optical branching cable is wrapped around a drum. According to the optical branching cable set described above, the extra length that extends along the curvature of the optical branching cable is formed. Therefore, by providing the sufficient extra length, the branching cord bag is less likely to fall off the optical branching cable.

Details of Embodiments of Present Disclosure

[0042] Specific examples of an optical branching cable and an optical branching cable set according to an embodiment of the present disclosure will be described below with reference to the drawings. The present disclosure is not limited to these exemplifications, but is indicated by the scope of claims, and is intended to include all changes within a scope and meaning equivalent to the scope of claims.

[0043] In the drawings, an arrow F indicates the forward direction of the illustrated structure. An arrow B indicates the rearward direction of the illustrated structure. An arrow U indicates the upward direction of the illustrated structure. An arrow D indicates the downward direction of the illustrated structure. An arrow R indicates the rightward direction of the illustrated structure. An arrow L indicates the leftward direction of the illustrated structure. These expressions related to the directions are used for convenience of description, and are not intended to limit the posture and direction of the illustrated structure in the actual use state.

Optical Branching Cable

[0044] FIG. **1** shows an optical communication system **1000** according to the present embodiment. As shown in FIG. **1**, the optical communication system **1000** includes a distribution board **1100** and a plurality of server racks **1200**. Optical branching cables **1** used in the optical communication system **1000** are collected and connected to the distribution board **1100**. The optical branching cable **1** is an example of an optical cable. The server rack **1200** accommodates a physical server therein. An optical fiber that is branched from the optical branching cable **1** is connected to each of the server racks **1200**. The optical branching cable **1** is laid on a cable laying tray T, and is wired to the distribution board **1100** and the server rack **1200**. The cable laying tray T is an example of a traction path.

[0045] FIG. **2** is a schematic view of the optical branching cable **1**. As shown in FIG. **2**, the optical branching cable **1** includes a main cable **10**, a branching cord portion **20**, a bend restricting component **40**, a distal end bag **50**, and a connection loop **60**.

[0046] FIG. **3** is a cross-sectional view of the main cable **10**. The main cable **10** includes a plurality of optical cords **11**, a tensile strength member **12**, and a tape **13**. The optical cord **11**, the tensile strength member **12**, and the tape **13** are provided along the longitudinal direction of the main cable **10**. The main cable **10** is also an example of an optical cable.

[0047] Each of the optical cords **11** includes at least one optical fiber **11**A therein. In the present embodiment, the optical cord **11** includes **24** optical fibers **11**A therein. The plurality of optical cords **11** are bundled together in a first bundle **14** to form an optical cord group **15**.

[0048] The tensile strength member **12** is made of fiber reinforced plastic (FRP) such as aramid FRP, glass FRP, or carbon FRP. The tensile strength member **12** is configured to share the tension when the optical branching cable **1** is pulled, so that excessive tension is not applied to the optical fiber **11**A. The optical cord group **15** and the tensile strength member **12** are bundled together in a second bundle **16** to form a core portion **17**.

[0049] The tape **13** forms the outermost layer of the main cable **10**. The tape **13** is longitudinally folded around the core portion **17** so as to maintain a shape that covers the core portion **17**. The tape **13** is a sheet-shaped insulating resin, and may be made of, for example, polyester.

[0050] The tape **13** is longitudinally folded such that a gap is formed between the tape **13** and the core portion **17**. Therefore, the outer diameter of the main cable **10** is configured to be contracted by an external force.

[0051] Returning to FIG. **2**, the branching cord portion **20** is a cord in which at least one optical fiber **11**A is led out from the optical cord **11** of the main cable **10** to the outside of the tape **13** which is the outermost layer. An optical connector **21** is provided at the distal end of the branching cord portion **20**. The branching cord portion **20** may be a portion in which the optical fiber **11**A in the optical cord **11** is led outward from the outermost layer of the main cable **10**, or the optical cord **11** itself may be led outward from the outermost layer of the main cable **10**.

[0052] The bend restricting component **40** is mounted at a position corresponding to a branching portion BR of the optical branching cable **1**. The bend restricting component **40** is provided at the branching portion BR such that bending with a radius of curvature smaller than a predetermined radius of curvature is less likely to occur in the branching cord portion **20** led out from the main cable **10**.

[0053] The distal end bag **50** protects the end portion of the optical cord **11** that is exposed at the distal end of the main cable **10**, and the end portion optical connector that is provided at the end portion of the optical cord **11**.

[0054] The connection loop **60** is connected to the tensile strength member **12** that is exposed at the distal end of the optical branching cable **1**. The connection loop **60** is a portion that is connected to a traction device such as a winch. The connection loop **60** is formed in a ring shape, and when the connection loop **60** is pulled, a tension is applied to the tensile strength member **12**, and the optical branching cable **1** can be displaced.

First Embodiment

[0055] Next, an optical branching cable set **100** according to the first embodiment will be described. FIG. **4** is a schematic diagram of the optical branching cable set **100** according to the first embodiment. As shown in FIG. **4**, the optical branching cable set **100** includes a plurality of optical branching cables **1** and a branching cord bag **110**. The plurality of optical branching cables **1** are arranged side by side in the depth direction of the paper in FIG. **4**.

[0056] The branching cord bag **110** is configured to collectively accommodate a plurality of branching cord portions **20** provided in the plurality of optical branching cables **1**. In the present embodiment, the branching cord portion **20** that is led out from the branching portion BR is wrapped in a coil shape and accommodated in the branching cord bag **110** together with the optical connector **21**.

[0057] The branching cord bag **110** may be made of a material having the cushioning property. The material having the cushioning property is, for example, a void cushioning material, a cloth material, or a resin material whose surface can be temporarily recessed.

[0058] The branching cord bag **110** may be fixed to the optical branching cable **1** at two different locations along the main cable **10**. In the present embodiment, the branching cord bag **110** is fixed to the main cable **10** by a hook-and-loop fastener. The branching cord bag **110** is provided with a first hook-and-loop fastener **110***a* and a second hook-and-loop fastener **110***b*. The first hook-and-loop fastener **110***a* and the second hook-and-loop fastener **110***b* form the hook surface of the hook-and-loop fastener. The first hook-and-loop fastener **110***a* is located in the leftward direction with respect to the second hook-and-loop fastener **110***b*.

[0059] The main cable **10** is provided with a third hook-and-loop fastener **10***a* and a fourth hook-and-loop fastener **10***b*. The third hook-and-loop fastener **10***a* and the fourth hook-and-loop fastener **10***b* form the loop surface of the hook-and-loop fastener. The third hook-and-loop fastener **10***a* is located in the leftward direction with respect to the fourth hook-and-loop fastener **10***b*. The third hook-and-loop fastener **10***a* and the fourth hook-and-loop fastener **10***b* may be formed on the main cable **10** of the plurality of optical branching cables **1** arranged in parallel.

[0060] The first hook-and-loop fastener **110***a* is configured to adhere to the third hook-and-loop fastener **10***a*. The second hook-and-loop fastener **110***b* is configured to adhere to the fourth hook-and-loop fastener **10***b*. The first hook-and-loop fastener **110***a* and the second hook-and-loop fastener **110***b* adhere to the third hook-and-loop fastener **10***a* and the fourth hook-and-loop fastener **10***b*, respectively, so that the branching cord bag **110** is fixed to the optical branching cable **1** at two different locations along the main cable **10**.

[0061] The branching cord bag **110** includes a base portion **111** and a top surface portion **112**. The base portion **111** is sandwiched between the main cable **10** and the branching cord portion **20** in a state where the branching cord portion **20** is accommodated in the branching cord bag **110**. The top surface portion **112** is located so as to cover the upper portion of the branching cord portion **20** in a state where the branching cord portion **20** is accommodated in the branching cord bag **110**.

[0062] The base portion **111** does not cover the branching portion BR to which the branching cord portion **20** is led out from the main cable **10**. On the other hand, the top surface portion **112** can cover the branching portion BR. When the dimension of the base portion **111** along the main cable **10** is L1 and the dimension of the top surface portion **112** along the main cable **10** is L2, the base portion **111** and the top surface portion **112** are formed so as to satisfy $L1<L2$. That is, the top surface portion **112** has a larger dimension along the axis of the main cable **10** than the base portion **111**.

[0063] The branching cord bag **110** includes an inner bag **113** configured to be accommodated inside the branching cord bag **110**. In the present embodiment, the branching cord bag **110** includes two inner bags **113**.

[0064] The inner bag **113** is configured to accommodate at least a part of the branching cord portion **20** of the optical branching cable **1**. In the present embodiment, all the branching cord portions **20** that are accommodated in the branching cord bag **110** are accommodated separately in the two inner bags **113**.

[0065] The inner bag **113** may be made of a material that is slippery relative to the inner surface of the branching cord bag **110**. For example, the inner bag **113** may be made of plastic. Accordingly, it is easy to perform the operation of storing and removing the inner bag **113** from the branching cord bag **110**.

[0066] The outer surface of the branching cord bag **110** may be formed with an extra length **114** that extends according to the curvature of the optical branching cable **1**. In the present embodiment, the extra length **114** is formed in a folded manner on the outer surface of the right side of the branching cord bag **110**. Alternatively, the extra length **114** may be formed on the outer surface of the left side of the branching cord bag **110**, as shown by the dashed line. The extra length **114** may be formed on both the outer surfaces of the left and right sides of the branching cord bag **110**.

[0067] FIG. **5** shows the curved optical branching cable **1** and the branching cord bag **110**. As shown in FIG. **5**, the optical branching cable **1** may be curved in the process of the operations such as manufacturing, transportation, and laying. For example, the optical branching cable **1** may be wrapped around a drum during manufacturing, and in this case, the optical branching cable set **100** is curved so as to be convex in the upward direction, which is the side on which the branching cord bag **110** is provided.
[0068] As the optical branching cable **1** is curved, the stress is applied to the branching cord bag **110** such that the branching cord bag **110** is stretched in the leftward direction and the rightward direction. In this case, the folded extra length **114** unfolds and extends along the curvature of the optical branching cable **1**, so that the branching cord bag **110** is less likely to be forcibly deformed, and the branching cord bag **110** is less likely to fall off the optical branching cable **1**.
[0069] According to the optical branching cable set **100** in the present embodiment, the branching cord bag **110** is configured to collectively accommodate the branching cord portions **20** of the plurality of optical branching cables **1**. Therefore, the plurality of branching cord portions **20** are protected by the branching cord bags **110**, the number of which is smaller than that of the optical branching cables **1**. Accordingly, since the number of components is reduced, it is possible to provide the optical branching cable set **100** that achieves both protection of the branching cord portion **20** and ease of manufacture.
[0070] Since the optical branching cable set **100** according to the present embodiment is made of a material having the cushioning property, the branching cord portion **20** is more effectively protected.
[0071] According to the optical branching cable set **100** in the present embodiment, the branching cord bag **110** is fixed to the plurality of optical branching cables **1** at two different locations along the main cable **10**. Therefore, the optical branching cable set **100** can be manufactured easily, and the branching cord bag **110** can be firmly fixed to the optical branching cable **1**.
[0072] According to the optical branching cable set **100** in the present embodiment, the base portion **111** does not cover the branching portion BR. Therefore, it is easy to perform the operation of accommodating the branching cord portion **20** led out from the branching portion BR. The top surface portion **112** is formed larger than the base portion **111** so as to cover the branching portion BR. Therefore, the branching cord portion **20** including the branching portion BR can be protected.
[0073] According to the optical branching cable set **100** in the present embodiment, since the inner bag **113** can be accommodated in the branching cord bag **110** and the branching cord portion **20** can be accommodated, the branching cord portion **20** is more effectively protected.
[0074] In the present embodiment, all the branching cord portions **20** that are accommodated in the branching cord bag **110** are accommodated separately in the two inner bags **113**. Accordingly, the branching cord portions **20** can be accommodated in a divided state in the branching cord bag **110** according to the connection place of the branching cord portion **20** and the necessary work. In particular, the work of connecting the plurality of branching cord portions **20** to the server rack **1200** is likely to be complicated since a large number of branching cord portions **20** are to be connected. Since the branching cord portions **20** are divided by the inner bag **113**, the laying work can be performed more efficiently.
[0075] According to the optical branching cable set **100** in the present embodiment, the extra length **114** that extends along the curvature of the optical branching cable **1** is formed in the branching cord bag **110**. Therefore, by providing the sufficient extra length **114**, the branching cord bag **110** is less likely to fall off the optical branching cable **1**.
Second Embodiment
[0076] Next, an optical branching cable set **200** according to a second embodiment will be described. FIG. **6** is a cross-sectional view taken along a plane orthogonal to the axis of the main cable **10** of the optical branching cable set **200**. Similarly to the optical branching cable set **100** according to the first embodiment, the optical branching cable set **200** includes the plurality of

optical branching cables **1** and a branching cord bag **210**. The branching cord bag **210** is configured to collectively accommodate the plurality of branching cord portions **20** provided in the plurality of optical branching cables **1**.

[0077] The branching cord bag **210** is mounted so as to be wrapped around the main cable **10** and the branching cord portion **20**. The branching cord bag **210** includes a first portion **211**, a second portion **212**, and a third portion **213**. The first portion **211** is located below the main cable **10**. The first portion **211** has a plurality of through holes H formed therein. The second portion **212** is sandwiched between the main cable **10** and the branching cord portion **20**. The third portion **213** covers the upper portion of the branching cord portion **20**.

[0078] The main cable **10** is fixed to the branching cord bag **210** by a bundling member **214**. The bundling member **214** is wrapped around the first portion **211** and the two main cables **10** through the through hole H. Accordingly, the main cable **10** is restricted from moving upward, forward, and rearward relative to the first portion **211**.

[0079] The second portion **212** and the third portion **213** adhere to each other by a hook-and-loop fastener. Accordingly, the space between the second portion **212** and the third portion **213** is closed, and the branching cord portion **20** is less likely to come out of the branching cord bag **210**.

[0080] Next, the assembly of the branching cord bag **210** will be described. The branching cord bag **210** is assembled through a first stage, a second stage, and a third stage. FIG. **7** shows the branching cord bag **210** in the first stage. The dashed line indicates the boundary between the first portion **211**, the second portion **212**, and the third portion **213**. The branching cord bag **210** in the first stage is not assembled with the first portion **211**, the second portion **212**, and the third portion **213**. The optical branching cable **1** is placed on the first portion **211**.

[0081] FIG. **8** shows the branching cord bag **210** in the second stage. As shown in FIG. **8**, in the second stage, the second portion **212** is folded onto the main cable **10**. Further, the branching cord portion **20** is provided on the second portion **212**. The second portion **212** has a slit S formed therein (see also FIG. **7**). The branching cord portion **20** that is led out from the main cable **10** in the branching portion BR is provided on the second portion **212** through the slit S.

[0082] The second portion **212** is provided with a fifth hook-and-loop fastener **212***a*. The fifth hook-and-loop fastener **212***a* faces upward when the second portion **212** is folded onto the main cable **10**. The fifth hook-and-loop fastener **212***a* forms the hook surface of the hook-and-loop fastener.

[0083] FIG. **9** shows the branching cord bag **210** in the third stage. As shown in FIG. **9**, in the third stage, the third portion **213** is folded over onto the branching cord portion **20**. Accordingly, the branching cord portion **20** is accommodated between the second portion **212** and the third portion **213**.

[0084] The third portion is provided with a sixth hook-and-loop fastener **213***a* (see also FIG. **8**). When the third portion **213** is folded onto the branching cord portion **20**, at least a part of the sixth hook-and-loop fastener **213***a* faces downward. The sixth hook-and-loop fastener **213***a* forms the loop surface of the hook-and-loop fastener.

[0085] In the third stage, the fifth hook-and-loop fastener **212***a* and the sixth hook-and-loop fastener **213***a* is configured to adhere to each other. The fifth hook-and-loop fastener **212***a* and the sixth hook-and-loop fastener **213***a* adhere to each other, so that the assembly of the branching cord bag **210** is completed.

[0086] According to the optical branching cable set **200** in the second embodiment, the branching cord bag **210** is configured to collectively accommodate the branching cord portions **20** of the plurality of optical branching cables **1**. Therefore, the plurality of branching cord portions **20** are protected by the branching cord bags **210**, the number of which is smaller than that of the optical branching cables **1**. Accordingly, since the number of components is reduced, it is possible to provide the optical branching cable set **200** that achieves both protection of the branching cord portion **20** and ease of manufacture.

Third Embodiment

[0087] Next, an optical branching cable set **300** according to a third embodiment will be described. FIG. **10** is a perspective view of the optical branching cable set **300** according to the third embodiment. FIG. **11** shows the optical branching cable set **300** and a branching cord bag **310** according to the third embodiment.

[0088] The optical branching cable set **300** according to the third embodiment includes 22 optical branching cables **1**. One branching cord bag **310** accommodates 22 branching cord portions **20**.

[0089] FIG. **12** shows an example of arrangement of branching cord portions **20** of the optical branching cables **1**. For the sake of description, the optical branching cables **1** will be referred to as a first optical branching cable, a second optical branching cable, . . . , and a twenty-second optical branching cable, in the order in which the optical branching cables are arranged from the top to the bottom on the paper surface of FIG. **12**. Similarly, the branching cord portions **20** provided in the first optical branching cable, the second optical branching cable, . . . , and the twenty-second optical branching cable will be referred to as a first branching cord portion **20**A, a second branching cord portion **20**B, . . . , and a twenty-second branching cord portion **20**V.

[0090] As shown in FIG. **12**, the branching cord portion **20** is divided into three rows in the longitudinal direction of the optical branching cable **1**. The branching cord portions **20** from the first branching cord portion **20**A to the seventh branching cord portion **20**G are provided in the upper row. The branching cord portions **20** from the eighth branching cord portion **20**H to the fourteenth branching cord portion **20**N are provided in the central row. The branching cord portions **20** from the fifteenth branching cord portion **200** to the twenty-second branching cord portion **20**V are provided in the lower row. Each branching cord portion **20** is wrapped clockwise in a coil shape in the plan view.

[0091] The branching cord portion **20** may be provided such that the bending radius is as large as possible. For example, in the upper row, the first branching cord portion **20**A is provided at a position farthest from the branching portion BR. The first branching cord portion **20**A to the seventh branching cord portion **20**G are arranged in this order at positions farther away from the branching portion BR. In other words, the seventh branching cord portion **20**G is provided at a position closest to the branching portion BR.

[0092] Similarly, in the central row, the eighth branching cord portion **20**H is provided at a position farthest from the branching portion BR. The eighth branching cord portion **20**H to the fourteenth branching cord portion **20**N are arranged in this order at positions farther away from the branching portion BR. In other words, the fourth branching cord portion **20**N is provided at a position closest to the branching portion BR. In the lower row, the fifteenth branching cord portion **20**O is provided at a position farthest from the branching portion BR. The fifteenth branching cord portion **20**O and the twenty-second branching cord portion **20**V are arranged in this order at positions farther away from the branching portion BR. In other words, the twenty-second branching cord portion **20**V is provided at a position closest to the branching portion BR.

[0093] By arranging the branching cord portions **20** in this manner, the branching cord portion **20** provided closest to the branching portion BR is the branching cord portion **20** of the rightmost optical branching cable **1** in each row. At this time, when the branching cord portion **20** is wrapped clockwise into a coil shape, the branching cord portion **20** provided closest to the branching portion BR is less likely to be bent with a bending radius smaller than a predetermined bending radius.

[0094] Since the branching cord portions **20** are also arranged in order when the branching cord portions **20** are removed, it is difficult to mix up the branching cord portions **20** when connecting the branching cord portions **20** to the server rack **1200** or the like, and the connection work of the branching cord portions **20** is easy.

[0095] According to the optical branching cable set **300** in the third embodiment, the branching cord bag **310** can collectively accommodate the branching cord portions **20** of the plurality of optical branching cables **1**. Therefore, the plurality of branching cord portions **20** are protected by the branching cord bags **310**, the number of which is smaller than that of the optical branching cables **1**. Accordingly, since the number of components is reduced, it is possible to provide the optical branching cable set **300** that achieves both protection of the branching cord portion **20** and ease of manufacture.

[0096] Although the present disclosure has been described in detail with reference to the specific embodiments, it is apparent to those skilled in the art that various changes and modifications can be made without departing from the spirit and scope of the present disclosure. In addition, the number, positions, shapes, and the like of the constituent members described above are not limited to those in the above embodiments, and can be changed to the numbers, positions, shapes, and the like suitable for carrying out the present disclosure.

[0097] The number of inner bags **113** provided in the branching cord bag **110** may be larger or smaller than two.

Claims

1. An optical branching cable set comprising: a plurality of optical branching cables; and a branching cord bag, wherein at least two of the optical branching cables each include: a main cable including: a plurality of optical cords each including at least one optical fiber; a tensile strength member; and a tape that forms an outermost layer; and a branching cord portion including at least one optical fiber that is led out from the optical cords of the main cable to an outside of the outermost layer, and wherein the branching cord bag is configured to collectively accommodate a plurality of branching cord portions provided in the plurality of optical branching cables.

2. The optical branching cable set according to claim 1, wherein the branching cord bag is made of a material having a cushioning property.

3. The optical branching cable set according to claim 1, wherein the branching cord bag is fixed to the plurality of optical branching cables at two different locations along the main cable.

4. The optical branching cable set according to claim 1, wherein the branching cord bag includes: a base portion that is sandwiched between the main cable and the branching cord portion in a state where the branching cord portion is accommodated in the branching cord bag; and a top surface portion that is located so as to cover an upper portion of the branching cord portion in a state where the branching cord portion is accommodated in the branching cord bag, wherein the base portion does not cover a branching portion to which the branching cord portion is led out from the main cable, and wherein the top surface portion has a larger dimension along an axis of the main cable than the base portion so as to cover the branching portion.

5. The optical branching cable set according to claim 1, wherein the branching cord bag includes an inner bag configured to be accommodated inside the branching cord bag, and wherein the inner bag is configured to accommodate at least a part of the branching cord portions of the optical branching cables.

6. The optical branching cable set according to claim 1, wherein an outer surface of the branching cord bag is formed with an extra length that extends according to a curvature of the optical branching cable.